

SNA Fall 2025: Assignment 1

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Assignment 1 Description

Libraries and Data

```
library(statnet)
load("luke.datasets.RData")
```

```
ls()
```

```
describe.Krebs()
```

```
class(Krebs)
```

```
## [1] "network"
```

```
summary(Krebs)
```

```
# View(Krebs)
```

```
network.size(Krebs)
```

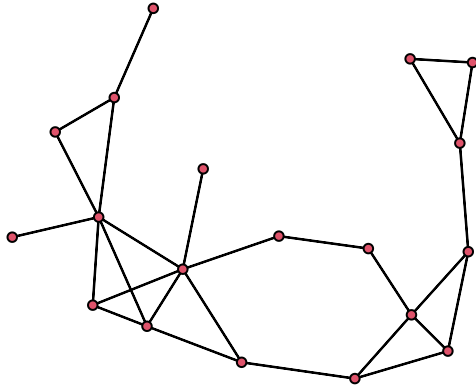
```
## [1] 19
```

```
network.edgecount(Krebs)
```

```
## [1] 27
```

```
set.seed(42)
```

```
plot(Krebs)
```



```
network::list.vertex.attributes(Krebs)
```

```
## [1] "flight"      "na"          "name"        "vertex.names"
```

```
network::get.vertex.attribute(Krebs, "flight")
```

```
## [1] "AA-77" "AA-77" "AA-77" "AA-77" "AA-77" "UA-93" "UA-93" "UA-93"
## [9] "UA-93" "UA-175" "UA-175" "UA-175" "UA-175" "UA-175" "AA-11" "AA-11"
## [17] "AA-11" "AA-11" "AA-11"
```

```
table(Krebs %v% "flight")
```

```
##
## AA-11 AA-77 UA-175 UA-93
##      5      5      5      4
```

```
network::list.edge.attributes(Krebs)
```

```
## [1] "na"
```

```
Krebs %e% "na"
```

```
## [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [13] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [25] FALSE FALSE FALSE
```

```
table(Krebs %e% "na")
```

```
##
## FALSE
##      27
```

```
Krebs_sociamatrix <- sna::as.sociomatrix.sna(Krebs)
vn <- network::get.vertex.attribute(Krebs, "name")
rownames(Krebs_sociamatrix) <- vn
colnames(Krebs_sociamatrix) <- vn
```

```
Krebs_matrix <- as.matrix(Krebs)
vn <- network::get.vertex.attribute(Krebs, "name")
rownames(Krebs_matrix) <- vn
colnames(Krebs_matrix) <- vn
dim(Krebs_matrix)
```

```
network::list.vertex.attributes(Krebs)
```

```
## [1] "flight"      "na"          "name"        "vertex.names"
```

```

nodes <- data.frame(
  id = 1:network::network.size(Krebs),
  name = network::get.vertex.attribute(Krebs, "name"),
  flight = Krebs %v% "flight"
)

write.csv(nodes, "Krebs_nodes.csv", row.names = FALSE)

network::list.edge.attributes(Krebs)

## [1] "na"

edges <- as.data.frame(network::as.edgelist(Krebs))
names(edges) <- c("from", "to")

write.csv(edges, "Krebs_edges.csv", row.names = FALSE)

```

Task 1: Paper and Network

1. Research question, context (what the study and network are about), main findings

1.1 Research question: To map and analyze a portion of the terrorist network centered on the 19 hijackers involved in the September 11th attacks, using publicly available data to visualize their interconnections and organizational structure.

1.2 Context: The study investigates how social network analysis (SNA) can uncover covert terrorist network patterns, helping researchers and investigators recognize key relational structures and signals when tracking such networks globally.

1.3 Main findings: The study reveals that investigators intentionally withheld or distorted network relationship data to mislead potential adversaries. This highlights the challenge of incomplete or manipulated data in network analysis. Moreover, measures of centrality are highly sensitive to even minor changes in connectivity, emphasizing the importance of data completeness and reliability.

1.3.1 Initially, Krebs (2002) examined prior trusted contacts (Erickson, 1981)—long-standing relationships built through shared living and learning experiences. When visualized through a network layout algorithm, these connections self-organized into the symbolic shape of a serpent, suggesting an organic, decentralized structure.

1.3.2 The network was highly sparse, and even members of the same flight team were separated by multiple relational steps—often more than two. This sparse structure minimized the risk of exposure if any member was captured or compromised. For a small network of fewer than 20 nodes, the average path length was 4.75, with some members over six steps apart. Such findings illustrate the trade-off between secrecy and efficiency in covert networks.

1.3.3 Through the temporary creation of transitory shortcuts (Watts, 1999), distant nodes coordinated efficiently across the network. Six such shortcuts reduced the average path length by over 40%, enhancing information flow and collaboration. Once coordination was achieved, these links became inactive. This dynamic illustrates how covert networks balance concealment with functionality (Baker & Faulkner, 1993).

1.3.4 The 19 hijackers operated within a broader support system of facilitators who did not board the planes. These collaborators acted as financial conduits and knowledge providers, supplying logistical, monetary, and technical resources essential to the operation.

1.3.5 Mohamed Atta exhibited the highest values across all centrality metrics—degree, closeness, and betweenness (Freeman, 1979)—indicating his strategic position as both the most active and the most influential node in the network, confirming his leadership role.

1.3.6 Covert networks that emphasize secrecy and stealth are extremely difficult to detect. However, by identifying patterns of limited yet periodic activation and targeting nodes with unique skills or high connectivity, investigators can gain valuable insights into the hidden relational dynamics of such networks.

2. Basic info about the networks, which you can obtain from R: number of vertices and edges (e.g., size); whether is directed or undirected; vertices and edges attributes; and any other details that you deem important

2.1 number of vertices and edges: The network consists of 19 vertices, representing the 19 hijackers involved in the September 11th attacks, and 27 edges that capture the relationships among them—such as shared training, cohabitation, schooling, and operational coordination (Krebs, 2002). The network was small and sparse, with an average path length of 4.75 steps and a clustering coefficient of 0.41.

2.2 whether directed or undirected The network is undirected, as the relationships between hijackers (such as training together or sharing living arrangements) were mutual and symmetric rather than one-way.

2.3 vertices and edges attributes: Vertices include personal attributes such as name and flight. Edges represent types of social ties—including strong ties (family or long-term friendship), moderate ties (co-training or co-living), and weak ties (single meeting or transaction). Additionally, six temporary “shortcut” edges were added to reflect coordinating meetings (Watts, 1999).

2.4 additional details: The network shows a “small-world” structure with a relatively high clustering coefficient and short average path length (The addition of six shortcuts reduced the average path length from 4.75 to 2.79, creating a relatively short path structure compared to the network’s size.), indicating that even in a covert network, members were connected through a few key links that enabled information flow while maintaining secrecy.

3. Paper’s strengths and weaknesses; one aspect you particularly liked and why; one aspect you did not like and why

3.1 Strengths and Weaknesses:

3.1.1 Strengths:

The paper makes a pioneering contribution by applying social network analysis (SNA) to visualize and interpret terrorist networks. By mapping the 19 hijackers’ social connections, Krebs (2002) provides a concrete framework for identifying hidden actors and understanding how covert networks operate. This analytical approach offers insights for preventing future terrorist attacks by revealing potential patterns of collaboration and coordination.

3.1.2 Weaknesses:

3.1.2.1 The study focuses on three centrality measures—degree, closeness, and betweenness—but omits indegree and outdegree, which are also crucial in directed networks. This is mainly because in-degree centrality reflects the extent to which an actor is the focus of attention or connection within a network. A high indegree indicates that many others in the network choose to connect with or rely on that actor, implying prestige, influence, or authority. In the context of a terrorist network, a member with high indegree centrality is likely to be a key leader or core figure who possesses critical information and coordinates important activities. While Krebs’s network is undirected, if it were modeled as a directed network, applying the concept of indegree could still help identify influential actors who attract more connections or attention from others. Including this perspective would have strengthened the leadership analysis of Mohamed Atta.

3.1.2.2 The paper successfully identifies small-world structures within the terrorist network but does not explore potential tensions or cooperation between these subnetworks. In reality, different small worlds may collaborate or compete, and both dynamics carry analytical significance. Cooperative ties can help investigators trace connections from one cell to another, whereas internal tensions or rivalries may lead to mistakes that expose critical information. Moreover, when relationships within each small world are strong but the overall network remains loosely connected, factions may form hidden alliances or pursue their own interests at the expense of the broader organization. Analyzing these inter-cell dynamics would deepen our

understanding of how internal fragmentation and conflict influence network stability and could offer strategic insights for disrupting terrorist organizations more effectively. However, the author does not address this point, leaving the discussion of inter-cell cooperation and conflict largely unexplored.

3.1.2.3 The paper provides limited discussion on the practical application of SNA in counter terrorism. While it demonstrates how networks can be mapped, it stops short of explaining how these findings can guide law enforcement in real investigations—from identifying recruiters and core members to disrupting communication chains. A deeper linkage between theory and practice would enhance its applied relevance.

3.1.2.4 The network model reflects terrorist communication patterns in 2001, yet organizational and technological contexts have since evolved. In the past, trust within terrorist networks was primarily built on geographic or kinship ties, but today it may also extend to members recruited and connected through online platforms. Likewise, tasks once relied on traditional channels such as phone calls and emails, whereas modern communication leaves extensive digital traces — including account metadata, chat logs, social interactions, IP addresses, activity timestamps, and device information. These persistent digital footprints can even amplify visibility, enabling researchers to reconstruct networks more accurately. Updating the framework to incorporate these new “digital edges” would improve its contemporary relevance.

3.1.2.5 Data limitations: The analysis relies solely on publicly available information, which likely excludes classified or unverified ties. As a result, the dataset represents only a partial view of the full terrorist network. This limitation affects the precision of relationship quantification and may cause bias in centrality interpretation.

3.2 Aspect I liked:

I particularly appreciated the author’s explanation of why covert terrorist networks are so difficult to predict or dismantle. Krebs clearly describes how strong ties among members remain mostly dormant and hidden, making detection nearly impossible. I also liked the two practical insights he proposed for identifying such networks: (1) detecting nodes that remain mostly silent but occasionally become active, and (2) targeting nodes that possess unique skills or resources. These ideas, when applied to modern contexts such as digital communication traces and online behavioral data, can help prevent terrorist activities more effectively.

3.3 Aspect I did not like:

The paper underemphasizes how SNA can be operationalized in counterterrorism efforts. While it effectively maps the network’s structure, it does not elaborate on how these analytical insights can guide real-world investigations—from identifying recruitment hubs to tracing communication flows. Expanding on this applied dimension would make the research more actionable for law enforcement and policy design.

Task 2: Core Network Measures (Define and Compute)

1. Density

Definition: Density is the proportion of observed ties (also called edges, arcs, or relations) in a network to the maximum number of possible ties. Thus, density is a ratio that can range from 0 to 1. The closer to 1 the density is, the more interconnected is the network (Luke, 2015, p.14).

Substantial definition: According to the lecture, density measures how cohesive or interconnected a network is. A higher density indicates that actors are tightly connected, which facilitates faster flow of information and resources, but also redundancy and constraint (Lecture 1 Slides, p.57). In covert terrorist networks, however, low density helps maintain secrecy and minimize exposure (Krebs, 2002). The 9/11 hijacker network’s density of 0.158 shows a deliberately sparse and compartmentalized structure.

Calculation:

$$D = \frac{2E}{N(N-1)} = \frac{2 \times 27}{19 \times (19-1)}$$

```
network::network.density(Krebs)
```

```
## [1] 0.1578947
```

2. Degree

Definition: In an undirected network, the degree d of a node i is given by the sum of the number of edges directly connecting node i with all other j nodes in the network (Lecture 1 Slides, p.58).

Substantial definition: Degree represents how active or well-connected a node is within the network. Nodes with higher degree occupy central positions and have greater access to information and influence. In the 9/11 hijacker network, Krebs (2002) shows that Mohamed Atta had multiple strong connections linking him to other key members, reflecting his leadership and coordination role (Krebs, 2002).

Calculation:

```
sna::degree(Krebs, gmode="graph")  
  
## [1] 1 3 2 1 6 3 4 3 3 1 6 2 2 4 3 3 3 2 2
```

I focus on Mohamed Atta because Krebs shows he occupies the core of the network in the neighborhood (expanded) graph, ranking first on degree-, betweenness-, and closeness-based centralities (Krebs, 2002, Figure 4 & Table 2). Mohamed Atta was the ring leader of this conspiracy (Krebs, 2002), and operated as a structural hole within the terrorism network. Consequently, removing high-degree nodes can generate huge disruption to terrorist organizations and therefore deserves attention.

```
deg <- sna::degree(Krebs, gmode="graph")  
names(deg) <- if ("name" %in% network::list.vertex.attributes(Krebs))  
  {network::get.vertex.attribute(Krebs, "name")}  
deg["Mohamed Atta"]  
  
## Mohamed Atta  
## 3
```

3. Degree Distribution

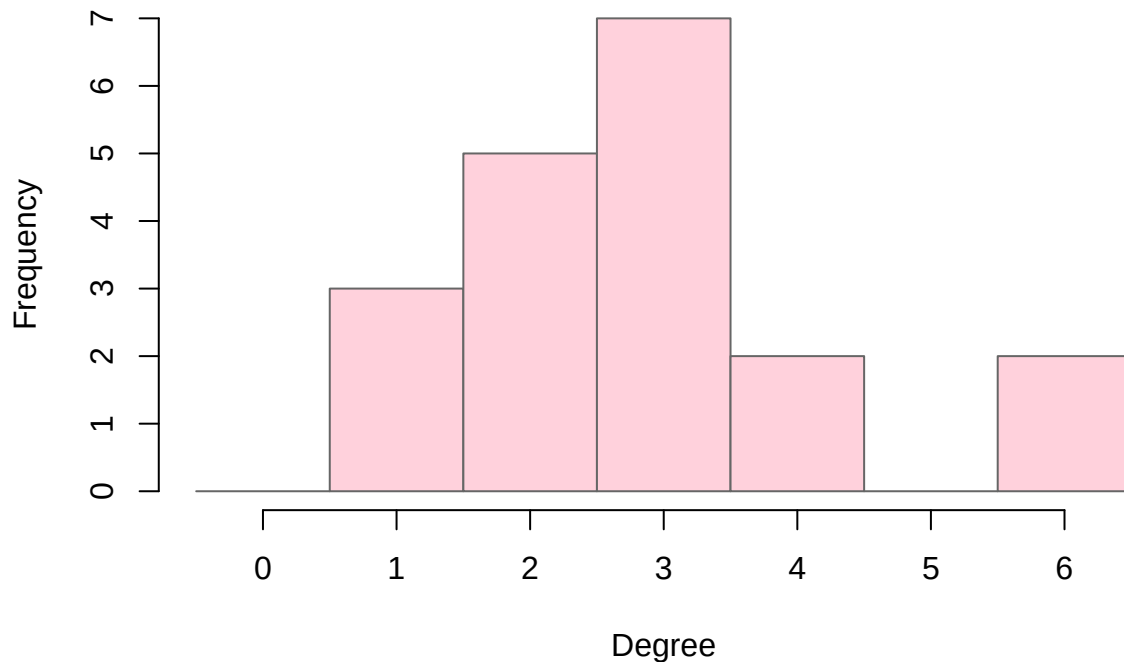
Definition: Frequency distribution of the degree values of all nodes in a network. It is a probability distribution describing the likelihood of a given degree in the network (Lecture 1 Slides, p.67).

Substantial definition: The distribution shows how connections are heterogeneously organized (e.g., whether most nodes have few ties and a few nodes have many ties). In covert contexts like Krebs's 9/11 network, a sparse structure with many low-degree nodes is consistent with secrecy and compartmentalization (Krebs, 2002, Figures 2–4).

Calculation:

```
deg <- sna::degree(Krebs, gmode="graph")  
breaks_seq <- seq(-0.5, max(deg) + 0.5, by = 1)  
  
hist(deg,  
      breaks = breaks_seq,  
      right = FALSE,  
      main = "Degree Distribution (Krebs network, 19 hijackers)",  
      xlab = "Degree",  
      ylab = "Frequency",  
      col = "#FFD1DC", border = "gray40"  
)
```

Degree Distribution (Krebs network, 19 hijackers)



Most nodes have low degree (e.g., 2–3), and only a few reach higher degrees (e.g., 4, 6); this skewed pattern reflects a sparse, compartmentalized network consistent with Krebs’s reconstruction of the hijacker network (Krebs, 2002, Figures 2–4).

4. Diameter

Definition: The diameter then for an entire network is the longest of the shortest paths across all pairs of nodes (Luke, 2015, p.15).

Substantial definition: The diameter indicates how “big” a network is in terms of steps needed to get from one side to the other, and it is often used as an upper bound for path lengths we analyze (Lecture 1 Slides, p.89) .

Calculation:

```
geo_dist = sna::geodist(Krebs, ignore.eval = TRUE)$gdist
diameter <- max(geo_dist[is.finite(geo_dist)])
diameter
```

```
## [1] 9
```

A diameter of 9 indicates that the shortest route from one side of the network to the other can require up to nine steps, implying a large path-length “span” and aligning with the sparse, compartmentalized structure described by Krebs (2002).

5. Shortest Paths

Definition: A path is the series of steps required to go from node A to node B in a network. The shortest path is the shortest number of steps required (Luke, 2015, p.15). And in weighted graphs, we need to inverse tie weights before applying the shortest path algorithm (Lecture 1 Slides, p.83).

Substantial definition: Shortest paths capture reachability and are the basis for measures like average path length and many centralities.

Calculation:

```
geo_dist <- sna::geodist(Krebs, ignore.eval = TRUE)$gdist

vn <- network::get.vertex.attribute(Krebs, "name")
atta_id <- match("Mohamed Atta", vn)

dist_from_atta <- geo_dist[atta_id, ]
mean_from_atta <- mean(dist_from_atta[is.finite(dist_from_atta)])

target_id <- which.max(dist_from_atta[is.finite(dist_from_atta)])
c(source = vn[atta_id], target = vn[target_id], distance = dist_from_atta[target_id])
```

```
##           source           target           distance
## "Mohamed Atta" "Majed Moqed"           "6"
```

From Atta, the average geodesic length = 3 (unweighted). His farthest reachable node is Majed Moqed at distance 6, while the diameter is 9.

We can know that distances from Mohamed Atta to others are generally shorter than the network's diameter, showing that he can reach most nodes in relatively few steps. This pattern supports Atta's central role observed in Krebs's reconstruction, even though his raw degree is modest in the 19-hijacker subgraph; many of his strategically relevant ties appear in the expanded neighborhood network (Krebs, 2002, Figures 2–4). This is the main reason why I choose Atta.

6. Clustering Coefficient

Definition: The tendency to form closed triangles (Luke, 2015, p.16).

Substantial definition: Higher clustering indicates tighter local cohesion and redundant ties, whereas lower clustering reflects sparse, compartmentalized structure. In Krebs (2002)'s constructed covert network, clustering within the cells is likely high, but cross-cell ties are sparse, leading to a moderate-to-low level of clustering.

Calculation:

$$T = \frac{\text{number of triangles}}{\text{number of triads}}$$

```
sna::gtrans(Krebs, mode = "graph")
```

```
## [1] 0.3970588
```

A global clustering (transitivity) of 0.397 means roughly 40% of connected triads are closed into triangles. This indicates non-trivial local cohesion within parts of the network, yet it remains far from fully clustered—consistent with a sparse, compartmentalized structure in a covert system (Krebs, 2002).

7.1 Additional Measure – Average Shortest Paths Length

Definition: Average shortest path length (ASPL) is the mean of all geodesic distances between node pairs in a connected network (Lecture 2 Slides, p.21, p.22).

Substantial definition: This metric captures the typical number of steps needed for information or influence to flow between any two actors. Shorter average paths imply faster potential communication or coordination. In Krebs (2002)'s terrorist network, a relatively short average path—despite overall sparsity—suggests a small-world-like structure: dense local cells connected by a few bridging ties (Krebs, 2002).

Calculation:

```
aspl <- mean(geo_dist[upper.tri(geo_dist) & is.finite(geo_dist)], na.rm = TRUE)
aspl
```



```
## [1] 3.51462
```

My aspl is 3.51, which falls between 2.79 (contacts + shortcuts) and 4.75 (contacts-only). This is mainly because Krebs's 4.75 corresponds to the contacts-only graph, and adding temporary shortcuts reduces the value to 2.79 (Krebs, 2002, Table 1). Thus, my result is reasonable. The average shortest path length of 3.51 indicates that, on average, information or coordination between hijackers would require about three to four steps. This aligns with the "small-world" pattern mentioned by Krebs (2002), where temporary shortcuts (e.g., shared training, meetings) reduce distances during active coordination, even though the overall graph remains sparse in ordinary times.

7.2 Additional Measure – Triad Census

Definition: Frequency of different patterns of triangles (Luke, 2015, p.180).

Substantial definition: By classifying triads according to open and closed triads, it helps us see the typical connection patterns among small groups of nodes.

Calculation:

```
tc <- sna::triad.census(Krebs, mode = "graph")
names(tc) <- c("003 (no ties)",
               "102 (one tie)",
               "201 (open triad, 2 ties)",
               "300 (closed triad, triangle)")

tc_prop <- tc / sum(tc)
round(tc_prop, 3)
```

```
##           0      1      2      3
## [1,] 0.587 0.361 0.042 0.009
```

```
T_from_triad <- tc["300 (closed triad, triangle)"] /
                (tc["201 (open triad, 2 ties)"] + tc["300 (closed triad, triangle)"])

T_from_triad
```

```
## 300 (closed triad, triangle)
##                               0.18
```

```
sna::gtrans(Krebs, mode = "graph")
```

```
## [1] 0.3970588
```

When using the function `triad.census()`, only 18% of all possible triads are fully closed ($T = 0.18$). However, when only considering connected triads using `gtrans()`, the global clustering coefficient rises to 0.397. This means that while most triads are unconnected, nearly 40% of those which are connected form closed triangles, indicating moderate local cohesion within cells.

8.1 Matrix Algebra Density

Definition and Substantial definition: the same as `l.density`.

Calculation:

```
n <- nrow(Krebs_matrix)
m <- sum(Krebs_matrix) / 2
density_matrix <- 2 * m / (n * (n - 1))
density_matrix
```

```
## [1] 0.1578947
```

The matrix-based density of 0.158 confirms a sparse network.

8.2 Matrix Algebra Degree

Definition and Substantial definition: the same as 2.degree.

Calculation:

```
degree_matrix <- rowSums(Krebs_matrix)
unnname(degree_matrix)

## [1] 1 3 2 1 6 3 4 3 3 1 6 2 2 4 3 3 3 2 2

vn <- network::get.vertex.attribute(Krebs, "name")
degree_matrix[match("Mohamed Atta", vn)]

## Mohamed Atta
## 3
```

Task 2: Core Network Measures (Interpret)

Interpretation of Measures

1. What does high or low density mean in this network? (For example, in the Krebs paper: What does density reveal about that terrorist network? Are terrorist networks typically high or low density, or it depends and why?)

A high density usually means a more connected and cohesive network. In Krebs (2002), the calculated density of 0.158 reflects a deliberately sparse and compartmentalized structure of the 9/11 terrorist network. Terrorist networks are typically relatively low density because such low density minimizes the risk of total network collapse if one cell is captured or compromised. Although such networks appear loosely organized and lack the strict hierarchical control typical of conventional organizations, individuals are nonetheless bound together by strong ideological commitment and shared purpose. This ideological cohesion mitigates the drawbacks of weak formal ties, enhances the network's adaptability, and facilitates the expansion and recruitment of additional individual actors. Therefore, this low-density structure represents a form of connection that warrants particular vigilance.

2. How does density relate to path length, degree distribution, or diameter in general and in your network? What does the relationship mean?

Low density, long average path length, and large diameter are consistent and together characterize a sparse and compartmentalized network structure. From the degree distribution, most nodes have low degree values (e.g., 2–3), which corresponds to a relatively large diameter.

In this context, I focus on the central node — Mohamed Atta. Although Atta's degree is relatively low (3), the shortest path from him to the farthest reachable node is only 6 steps, smaller than the network's diameter of 9. This indicates that he can reach most other nodes through only a few steps. While degree measures only a node's direct connections, it does not capture its control or influence over neighboring nodes.

Thus, even though Atta's own degree is not high, he exerts influence by controlling several neighboring nodes with higher degrees, which amplifies his importance within the network. According to Table 2 in Krebs (2002), Atta ranks first in both Betweenness and Closeness centralities, confirming his leadership role in the network.

3. Is transitivity high or low in this network? What does it mean?

The overall transitivity (global clustering coefficient) of this network is moderate ($T = 0.397$). This means that roughly 40% of all possible triads are closed into triangles, indicating that some local groups of hijackers are tightly interconnected while others remain isolated. Such a moderate level of transitivity reflects a partially cohesive but deliberately compartmentalized network structure. In Krebs (2002), this pattern is consistent with the cell-based design of the 9/11 network — high clustering within individual cells (e.g., flight teams) but few links across cells.

This structure has two main advantages for terrorist organization (though of course it doesn't work for us):

First, within small groups (small world) characterized by a high clustering coefficient, frequent interactions among members strengthen mutual trust, forming strong ties. In turn, these strong ties further reinforce the organization's terrorist narratives, deepening emotional bonds and mutual trust among individual actors, thereby making the internal connections even more resilient.

Second, at the overall sparse network level, the structure not only reduces the risk of detection but also facilitates the wider dissemination of extremist narratives, gradually drawing more individuals into the organization's web of weak ties, some of whom may eventually become active nodes within the terrorist network itself.

4. Are paths between nodes generally short or long? What does this imply for your network?

According to Krebs, for a small network of less than 20 nodes, we can see a long average path length of 4.75 steps. Several of the hijackers are separated by more than 6 steps (Krebs, 2002, table 1). So in terms of "contacts-only", paths between nodes are generally long. But it can be reduced to 2.79 with some "shortcuts" (e.g. meetings). So overall, the paths between nodes are moderate-to-long: information can spread relatively efficiently, but not as quickly as in a fully connected small-world structure. This result indicates that the 9/11 hijacker network exhibits a semi-small-world pattern, some transitory "shortcuts" shorten paths, but overall the network remains deliberately sparse and compartmentalized. So it really depends.

5. What does the triad census (if you calculated it) reveal about your network?

In my network, the triad census is only 0.18 and the clustering coefficient (transitivity) is 0.397, meaning roughly 40% of all triads are closed. This indicates that while some local groups are strongly interconnected, a majority of triads remain open, suggesting a moderately cohesive but largely sparse network structure. This balance between closure and openness reflects a strategic trade-off between efficiency and secrecy closed triads enhance trust to improve coordination efficiency and open triads helps reduce the risk of exposure

Closed triads often form among members of highly homogeneous groups, where people share similar backgrounds or experiences. As these members interact more frequently, trust and emotional bonds grow stronger. Because of this solid foundation of trust, shared knowledge, and common understanding, such groups can coordinate efficiently when the network is activated.

However, too many closed triads can create several problems: First, they lead to redundancy and higher exposure risk. If one person in this terrorism network is caught, the whole cluster can be revealed. Second, they limit diversity and outreach, making it harder for the terrorism organization to spread its ideology to outsiders. Finally, maintaining closed triads is costly in terms of time, energy, and money, so terrorist networks can usually sustain only a few closed triads, while most ties remain open.

This pattern, as Krebs (2002) describes, is "sparse but cohesive" — locally dense but globally sparse. It allows the network to stay resilient and adaptable under threat, while also fitting the small-world idea that a few bridging ties can keep the network connected efficiently.

Thus the triad census shows that the hijacker network was neither purely fragmented nor fully cohesive, but strategically organized around local trust clusters connected through minimal bridging ties, a hybrid structure that maximizes both coordination and secrecy.

Assessing the fit of Theories to the Network

1. Small-World

Author Luke [A User's Guide to Network Analysis in R, page 152] argues that [small-world model networks have more realistic levels of transitivity along with small diameters]. I argue that the 9/11 hijacker network constructed by Krebs (2002, p.4, Table 1) does exhibit [small-world] features because the transitivity of 0.397 indicates moderate local cohesion, while the average path length of 3.51 (between the contact-only and contact + shortcut versions) suggests relatively short global distances. Together, these metrics imply dense local cells connected by a few bridging ties — a defining feature of small-world structures. Moreover, the specific context of the paper [e.g., terrorist networks] suggests, temporary "shortcuts" such as joint meetings

reduced distances between cells from 4.75 to 2.79, facilitating coordination while maintaining secrecy. Thus, the network fits the small-world pattern: locally dense but globally sparse, allowing both coordination and concealment.

2. Scale-Free

Author Luke [A User's Guide to Network Analysis in R, page 155] argues that [Numerous studies, in fact, have shown that a wide variety of observed networks have heavy-tailed degree distributions that approximately follow a power law. These are typically called scale-free networks]. That means scale-free networks should follow a power-law degree distribution, where a few hubs possess many connections and most nodes have very few connections. I argue that this network does not exhibit [scale-free] features because the degree distribution from my calculation (see histogram) shows that even though most nodes have low degrees (2–3) and only two have relatively high degrees (6), no node displays the extreme connectivity typical of hubs. That means the degree distribution is only slight heterogeneity, in other words, a slightly skewed small network rather than a truly scale-free structure. Moreover, its slight skewness is likely the result of intentional design rather than natural self-organized hub formation: as Krebs (2002) explains, covert terrorist networks trade efficiency for secrecy, deliberately constraining their connections to avoid detection.

3. Balance Theory

Author Wasserman, S., & Faust, K. (1994)[Social Network Analysis: Methods and Applications, page 220] argues that triadic balance occurs when friends of friends tend to be friends, forming stable clusters. The clustering coefficient (transitivity = 0.397) reveals that about 40% of triads are closed, indicating moderate balance within cells but not across them. This supports Krebs (2002), who observed that hijacker teams were internally cohesive but links across cells were minimized to avoid exposure. Therefore, the network shows a state of localized balance but global imbalance.

In summary, the 9/11 hijacker network constructed by Krebs is a small-world but not scale-free system, exhibiting partial balance within clusters but not across them.

Reflections & AI Use

Learning Reflections

I think this task is challenging in several ways:

First, for beginners, it's a little difficult to adapt and modify the code from Lab 1 for new applications.

Second, when it comes to explaining the concepts, there are no fixed reference values to compare with. For example, it's hard to tell exactly how long a path length or diameter should be to be considered "long", or how high a transitivity value indicates strong cohesion. Sometimes we rely on our structural intuition, but sometimes that intuition could be wrong, so this part requires more experience and accumulation to make accurate judgments.

Finally, many of the concepts overlap. For instance, transitivity in Task 2 and the triad census measure similar ideas, as do small-world and balance theory in the assessment. I find it a bit difficult to clearly explain the differences between them.

AI Disclosure Statement

I used ChatGPT to help me revise and debug my code.

For example, I set the matrix row and column names with `rownames(Krebs_sociamatrix) <- vn`

```
got vertex name with names(deg) <- if ("name" %in% network::list.vertex.attributes(Krebs))
{network::get.vertex.attribute(Krebs, "name")}
```

drew the degree histogram with `hist`

Atta's node by name using `atta_id <- match("Mohamed Atta", vn)` .

References

A User's Guide to Network Analysis in R

Uncloaking Terrorist Networks

Chapter 6 Structural Balance and Transitivity, Structural Balance and Transitivity

Chapter 14 Triads, Social Network Analysis: Methods and Applications